

Virtual Experts to Collective Decisions: An LLM-Driven Majority Voting for EM

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Abstract— Entity Matching (EM) — the task of identifying records that refer to the same real-world entity across heterogeneous data sources — remains a critical challenge in data integration and deduplication. While traditional approaches rely on rule-based systems or supervised learning models, recent advances in large language models (LLMs) offer new paradigms for decision-making in EM tasks. In this work, we present an experimental framework where a GPT-based model simulates a panel of three virtual domain experts. Each expert independently evaluates whether a candidate pair refers to the same entity, and the final decision is determined through a majority-vote mechanism. This study examines the feasibility and effectiveness of such simulated expert panels in EM, providing insights into their decision consistency, error patterns, and potential integration into broader EM pipelines.

Research Keywords— Entity Matching, LLMs, Natural Language Processing

1 INTRODUCTION

Entity Matching (EM) the task of identifying records that refer to the same real-world entity across different data sources [1] has become increasingly important in domains such as e-commerce, healthcare, and digital libraries. In today's data-driven world, accurate and efficient EM is crucial for applications like personalized recommendations, patient data integration, and knowledge graph construction. Traditional EM approaches often rely on manual feature engineering, heuristic rules, or string similarity measures, which struggle to generalize across domains and data formats. These approaches require considerable human intervention and makes errors when applied to unseen or heterogeneous datasets.

With the rise of Transformer-based models [2], particularly GPT model [3]. New opportunities have emerged to solve EM using contextualized, deep semantic representations. GPT models offer strong capabilities in natural language generation, zero-shot reasoning, and data augmentation. The generative capabilities of GPT models can be applied to EM pipeline, thereby enhancing their robustness and ability to generalize to new domains. Unlike traditional approaches that rely on manually crafted rules, GPT's ability to generate text allows it to enhance EM pipelines through tasks such as data augmentation, candidate filtering, and even end-to-end matching. We propose a novel prompt format based on research on EM pipeline using GPT. This demonstrates how GPT models excel in situations with limited labeled data. Furthermore, we demonstrate how GPT's generative and augmentation capabilities can enhance the pipeline, enhancing the accuracy and scalability of entity analysis across diverse data sources.

2. RELATED WORK

Traditional EM Methods: Early approaches to EM include rule-based systems[4], similarity scoring (e.g., Levenshtein distance, Jaccard index), and classical probability based models[5] using handcrafted features. These systems often require extensive domain knowledge and are sensitive to noise and schema variations.

Neural Network-Based EM : Recent advances include deep learning models such as Siamese networks, which compute vector representations of entity pairs for similarity estimation. Embedding-based methods using static word vectors (Word2Vec[6], GloVe) have shown improvements, but lack contextual sensitivity[7].

PLM-Based EM : Using Pretrained Language models (PLMs) such as BERT and RoBERTa, leverage contextualized word representations for EM[8]. These models excel in tasks requiring deep semantic understanding, particularly when fine-tuned on domain-specific data[9]. PLM-based approaches are effective in comparing entity pairs by considering the context of words, but they require significant labeled data for fine-tuning and may struggle with unseen entities. Despite these challenges, PLMs provide a solid foundation for EM tasks when ample domain data is available.

3. METHODOLOGY

To mitigate single-prompt fragility and improve decision reliability in entity matching, we adopt a *committee prompting* scheme in which the model is instructed to simulate three independent reviewers (Expert A, B, C). Each expert processes the same pair of records in isolation and produces (i) a one-sentence rationale that highlights the most diagnostic evidence and (ii) a binary vote (YES/NO) on whether the records denote the same real-world entity. The responses are aggregated by a deterministic majority rule—if at least two experts vote YES, the *Final Decision* is YES; otherwise NO. Independence is encouraged through explicit instructions that prohibit cross-referencing among experts, and parsimony is enforced by capping rationales at one sen-

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tence to bound token cost. We also standardize the output via a strict schema (fixed field names and uppercase labels) to enable lossless, programmatic parsing.

This design offers three methodological advantages. First, by eliciting multiple reasoning paths from a single model, it behaves like a lightweight mixture-of-experts and reduces variance from idiosyncratic prompt interpretations. Second, the short, attribute-grounded rationales yield auditability—errors can be traced to specific evidence—without revealing long chain-of-thought. Third, the aggregation and schema constraints make the method easy to integrate into automated pipelines (e.g., regex or JSON parsing) while keeping inference overhead modest relative to few-shot exemplars.

4. EXPERIMENTS

We conducted our experiments using 1,000 randomly sampled test instances from the Amazon-Google and DBLP-Scholar subsets of the ER_Magellan dataset, a widely used benchmark for entity matching (EM). When using prompts that mimic a voting process, we observed an F1-score of 0.970 on the DBLP-Scholar dataset and 0.692 on the Amazon-Google dataset.

5. CONCLUSION

The proposed prompt, designed to emulate a voting mechanism, demonstrated approximately a 2-percentage-point improvement in accuracy within the academic/library domain when compared to prior zero-shot prompting strategies. While further performance gains may be achievable through variations of the prompt structure or by incorporating few-shot examples, such directions are beyond the scope of this study and are left for future research.

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